



Pablo Martínez Agulló

Senior AI Engineer
PhD in Physics

Senior AI Engineer specialised in agentic systems: multi-agent architectures, LLM orchestration and RAG, taken from prototype to production. At Indra's innovation lab I design planning and decision-support agents for defence and commercial products (TRL 2–5). Previously shipped GenAI products at Mercadona on GCP. PhD in Physics (CERN/ATLAS) behind the engineering: large-scale data, ML and statistical rigour.

EDUCATION

2018 - 2024

Doctorate in Physics | University of Valencia

2016 - 2017

MSc in Advanced Physics | University of Valencia

2015 - 2016

Erasmus | RWTH Aachen University

2011 - 2015

Degree in Physics | University of Valencia

EXPERIENCE

September 2025 - Present

Indra | Senior AI Engineer (Spain)

- Multi-agent GenAI systems for defence and commercial products (TRL 2–5)
- Agents flying real drones from natural-language mission orders, on a local LLM
- Planning, metacognitive and decision-support agents
- Custom MCP servers and A2A agent orchestration
- QUBO/Ising planning solved with quantum annealing
- Secured an externally funded CSIC–Indra research collaboration
- Agent observability with LangSmith and OpenTelemetry

September 2024 - September 2025

Mercadona | Artificial Intelligence Engineer (Spain)

- RAG assistant over internal policy docs: LangChain, Gemini, FAISS, Dialogflow CX
- Structured extraction from product-label images: Gemini vision, Kafka, SQL
- Production on GCP and Vertex AI: FastAPI, Docker, Kubernetes, CI/CD
- Workforce scheduling optimiser with Pyomo and Gurobi

December 2017 – May 2024

IFIC/CERN | Researcher (Spain/Switzerland)

- Member of the ATLAS collaboration
- Petabyte-scale data analysis with supervised ML (BDT, neural networks)
- Full-stack detector-alignment monitoring app: adopted across the 5,500-member collaboration, in production since 2019
- Shared analysis software in C++ and Python (ATLAS Athena)
- PhD thesis on the Higgs–top interaction; JHEP 2022 and 2025

February 2020 – September 2021

University of Valencia | Lecturer (Spain)

- Electromagnetism and Physics Laboratory for the Chemistry degree: 100+ students

HARD SKILLS

Agentic AI: LangGraph, CrewAI, AutoGen, OpenAI Agents SDK. Single-agent and multi-agent architectures, planning and metacognitive agents, evaluator–optimizer and human-in-the-loop patterns, custom MCP servers, A2A protocol, agent tooling and routing.

LLM engineering & adaptation: prompt and context engineering, structured outputs and schema validation (Pydantic), tool/function calling, guardrails, model selection and benchmarking, evaluation-driven iteration.

RAG & document intelligence: chunking and embedding strategies, FAISS, hybrid retrieval and reranking, OCR with LLMs, structured extraction from images and PDFs, Dialogflow CX.

Edge & on-prem AI: local model serving (Ollama), constrained and air-gapped deployments, MQTT/Kafka event pipelines, FPGA and High-Level Synthesis fundamentals.

Evaluation & observability: LangSmith, OpenTelemetry, OpenObserve, offline eval sets, LLM-as-judge.

Platform & MLOps: FastAPI, Docker, Kubernetes, GCP (Vertex AI, Cloud Functions, Cloud Storage), CI/CD (Spinnaker, CloudBees), hexagonal architecture, CLEAN/SOLID.

Optimisation: QUBO/Ising formulations, quantum annealing (D-Wave), mixed-integer programming with Pyomo and Gurobi, scheduling and resource allocation.

ML & data: PyTorch, TensorFlow/Keras, XGBoost, scikit-learn, pandas, NumPy, SciPy, statistical modelling, feature engineering.

Programming: Python, C++, SQL, Bash, R.

OUTREACH

- Speaker and host of the event *70 años del CERN* (800+ attendees, 2025)
- Contributor to the physics podcast *Oscilador Armónico* (2023 - 2024)
- Speaker/host of the public event *10º aniversario del descubrimiento del bosón de Higgs* (700+ attendees, 2022)
- High school talks: *¿De qué está compuesta la realidad?* (2022 - 2023)
- Video *LHCRun3* (21k+ views, 2022)
- Organizer of the science fairs *Expociencia* and *Experimenta* (2019 - 2022)

CONTACT



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GitHub: [@MartinezAgullo](#)

SCHOLARSHIPS

- 2019: Scholarship from the Valencian Government (GVA) for the hiring of research personnel
- 2017: Severo Ochoa Collaboration Scholarships at CSIC Centers of Excellence at IFIC

SOFT SKILLS

- Collaboration & cross-functional teamwork
- Autonomy
- Leadership
- Strong communication and interpersonal skills
- Creative problem-solving
- Extensive experience in international environments
- Proactive & eager to learn
- Time and deadline management
- Critical & strategic thinking

LANGUAGES

- Spanish :: Native
- English :: Professional
- Catalan :: Native
- German :: Basic